FYP PROJECT PRESENTATION: ENVIRONMENT AND SUSTAINABILITY (PANEL)

PROGRAMME :	DIPLOMA IN BIOMEDICAL ELECTRONICS ENGINEERING (DIAGNOSTIC/ICT/LABORATORY/THERAPEUTIC/RADIOLOGY)			**DURATION:**
TOPIC:	FYP PROJECT PRESENTATION: ENVIRONMENT AND SUSTAINABILITY (PANEL)			WEEK 7
COURSE:	DBD/DBI/DBL/DBT/DBR - 51014 FINAL YEAR PROJECT			
RELATED CLO:	CLO3 - Understand the impact of engineering technology solutions in societal and environmental context and deminstrate knowledge of and need for sustainable development. (PLO6, A4, DK7)			
SEM / SESSION :	5 / JUL - DEC 2026	**CLASS:**		**WEIGHTAGE:**
STUDENT 1 NAME :		**MATRIX ID:**		10%

LO7 ENVIRONMENTAL AND SUSTAINABILITY

No.	DOMAIN (ETAC 2024)	DOMAIN (ETAC 2020)		ELEMENT ENVIRONMENT AND SUSTAINABILITY	Level	Markah	Factor / FP	Basic Mark	Given Mark
1	CLO3: PLO6, A4, DK7	CLO3: PLO7, A4, DK7	6.4-A	**Application of Environmental Sustainability Concepts**	The need for life to depend directly or indirectly towards environment. Student should be able to: 1. Understand the concepts of environmental. 2. Have awareness toward environmental and sustainability. 3. Relate knowledge learned with environmental and sustainability 4. Suggest new activites relate with environmental and sustainability concept. 5. Choose an implementation plan that produce the best possible outcome toward environmental sustainability .				
				Comply with any 1 from the criteria listed.	Weak	1	1.20	1.20	
				Comply with any 2 from the criteria listed.	Poor	2		2.40	
				Comply with any 3 from the criteria listed.	Fair	3		3.60	
				Comply with any 4 from the criteria listed.	Good	4		4.80	
				Comply with all criteria listed	Excellent	5		6.00	
			6.4-B	**Application of Sustainability to the Environment**	Attempt to preserved environment, natuere and minimise negative effect from human activity. Student should be able to: 1. Include process/ product/ service toward green technology. 2. Implement safe process and material that enviromental friendly. 3. Handle any residue or waste systematically. 4. Reduce wastage of material or energy. 5. Handle any work proses with consideration of environmental sustainability. 6. Use any renewable energy. 7. Use any recyle material. 8. Do environmental impact analysis.				
				Comply with any 2 from the criteria listed.	Weak	1	1.20	1.20	
				Comply with any 3 from the criteria listed.	Poor	2		2.40	
				Comply with any 4 from the criteria listed.	Fair	3		3.60	
				Comply with any 5 from the criteria listed.	Good	4		4.80	
				Comply with no less than 6 criteria listed	Excellent	5		6.00	
			6.4-C	**Innovation in Environmental Solutions**	This element focuses on the student's ability to generate innovative solutions that address environmental challenges by integrating creativity, technology, and sustainable practices. Student should be able to: 1. Develop innovative ideas/products that reduce environmental impact. 2. Apply sustainable design thinking in engineering or project solutions. 3. Integrate renewable energy or sustainable resources into practical applications. 4. Use digital tools, modelling, or simulation software to evaluate environmental performance. 5. Create prototypes or demonstrations that showcase innovative approaches to sustainability. 6. Present innovative solutions clearly with supporting evidence or data. 7. Evaluate the effectiveness of innovative solutions based on environmental, social, and economic outcomes.				
				Comply with any 2 from the criteria listed.	Weak	1	0.90	1.20	
				Comply with any 3 from the criteria listed.	Poor	2		1.80	
				Comply with any 4 from the criteria listed.	Fair	3		2.70	
				Comply with any 5 from the criteria listed.	Good	4		3.60	
				Comply with no less than 6 criteria listed	Excellent	5		4.50	
			6.4-D	**Environmental Awareness & Social Responsibility**	This element emphasizes the student's role in understanding environmental issues and taking responsibility to promote sustainability within the community, industry, and society. Student should be able to: 1. Demonstrate awareness of environmental regulations, policies, and standards. 2. Participate actively in programs or activities that promote environmental sustainability. 3. Encourage peers or community members to adopt sustainable practices. 4. Evaluate the long-term social, cultural, and economic impacts of environmental sustainability. 5. Lead or organize environmental campaigns, projects, or green initiatives. 6. Collaborate with stakeholders (community, industry, government) to support sustainability practices. 7. Reflect on personal and collective responsibility in protecting the environment.				
				Comply with any 2 from the criteria listed.	Weak	1	0.90	1.20	
				Comply with any 3 from the criteria listed.	Poor	2		1.80	
				Comply with any 4 from the criteria listed.	Fair	3		2.70	
				Comply with any 5 from the criteria listed.	Good	4		3.60	
				Comply with no less than 6 criteria listed	Excellent	5		4.50	

			6.4-E	**Safety, Ethics & Regulatory Compliance in Sustainable Practice**					
				Ignores safety, ethical responsibility, and regulatory considerations.	Weak	1	0.90	1.20	
				Limited attention to safety, ethics, or compliance; potential risks not adequately addressed.	Poor	2		1.80	
				Basic safety and ethical considerations applied, but explanations lack depth or consistency.	Fair	3		2.70	
				Complies with most safety and ethical requirements; shows clear awareness of environmental and social responsibility.	Good	4		3.60	
				Fully complies with safety standards, ethical practices, and relevant regulations. Clearly explains environmental, social, and safety implications of the project.	Excellent	5		4.50	
			6.4-F	**Sustainable Material Selection & Resource Efficiency**					
				No consideration of sustainability, efficiency, or resource impact in material/component selection.	Weak	1	0.90	1.20	
				Minimal consideration of sustainability in material selection; choices are mostly convenience-based.	Poor	2		1.80	
				Uses standard materials with some consideration of sustainability, but justification is limited or qualitative only.	Fair	3		2.70	
				Appropriate sustainable materials or components selected with reasonable justification. Shows awareness of efficiency and waste reduction.	Good	4		3.60	
				Material/component selection is clearly justified based on sustainability, durability, efficiency, and lifecycle impact. Demonstrates measurable reduction in waste, energy, or resource use.	Excellent	5		4.50	
Obtain Marks						30	**6**	**30**	**0**
Total Marks (100%)									**0.00**

VERIFICATION

SIGNATURE & OFFICIAL STAMP:

DATE: